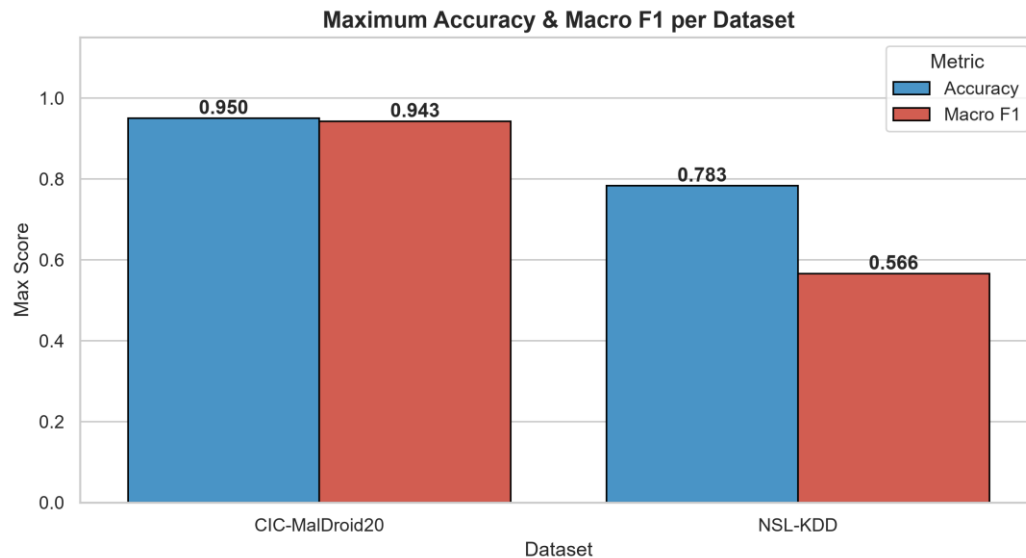
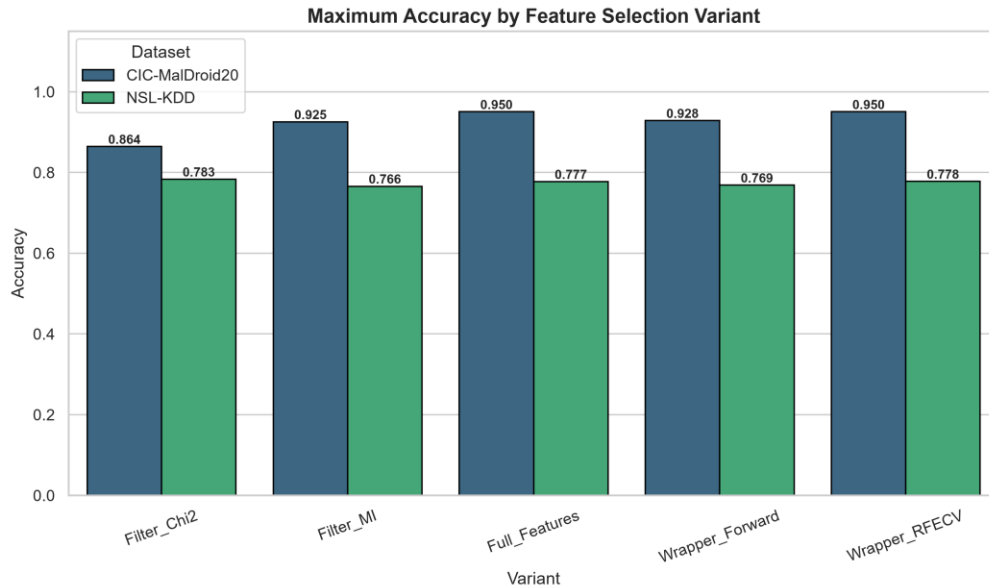


Analysis Requirements: Cross-Dataset Comparative Report



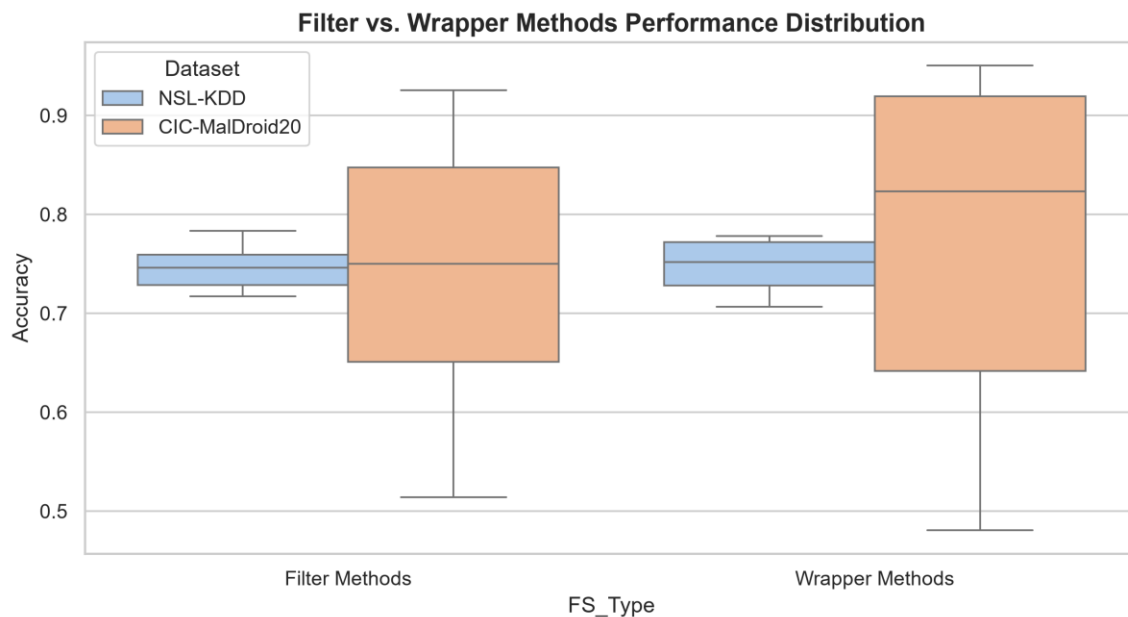
Analysis :

CIC-MalDroid20 gives significantly better performance (Accuracy > 95%). Why? NSL-KDD suffers from severe class imbalance (e.g., U2R attacks are extremely rare) and a well-known distribution shift where test data contains attack types unseen in training data. CIC-MalDroid20, despite having more features, is more balanced and homogeneous, allowing models to easily find clear patterns.



Analysis:

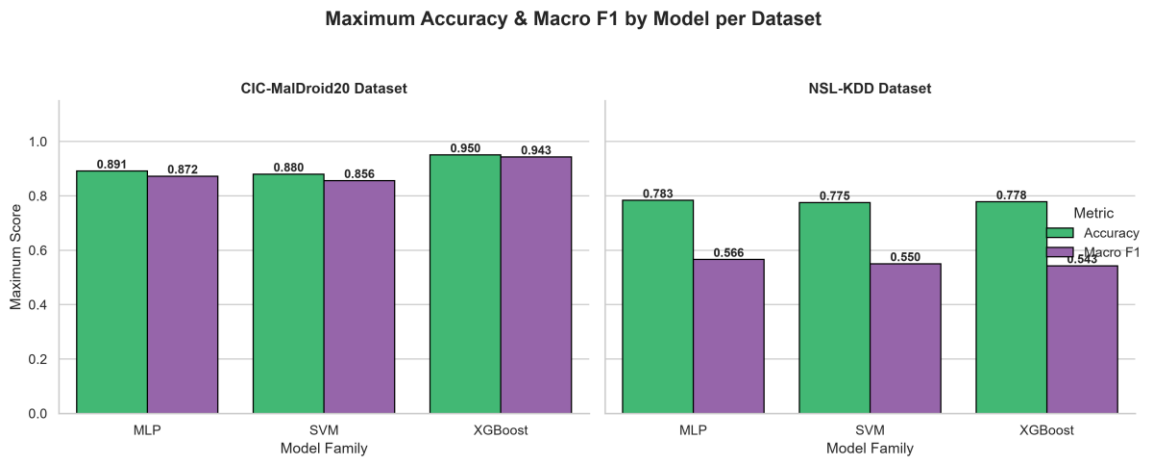
Feature selection does NOT consistently improve absolute Accuracy. For CIC-MalDroid20, 'Full Features' yielded the highest accuracy (95.04%). For NSL-KDD, specific methods like Filter_Chi2 slightly improved MLP's performance. The primary benefit of feature selection in our experiment was drastically reducing dimensionality and prediction time rather than boosting accuracy.



Analysis:

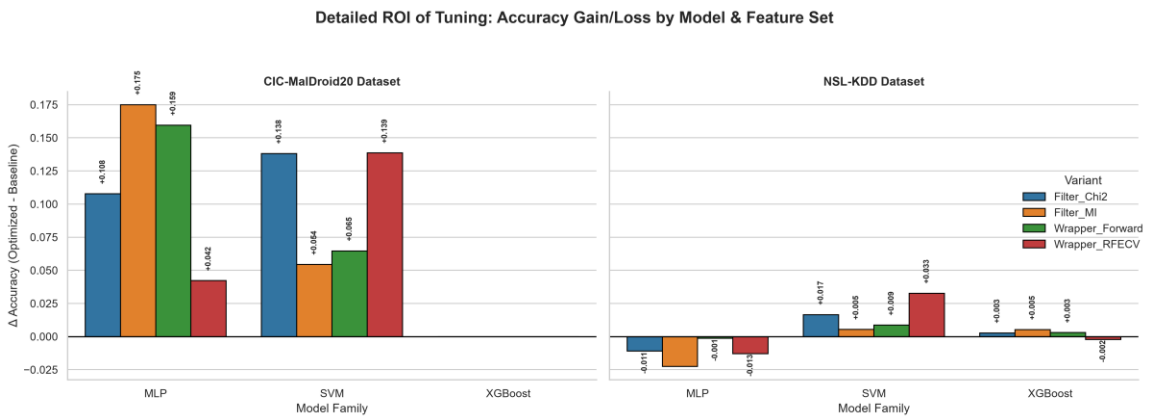
Wrapper methods (like RFECV) generally outperformed Filter methods because they use the actual machine learning model (e.g., XGBoost) to evaluate feature combinations,

maintaining accuracy closer to the 'Full Features' baseline. Filter methods are much faster but rely on generic statistical metrics, leading to occasional performance drops.



Analysis:

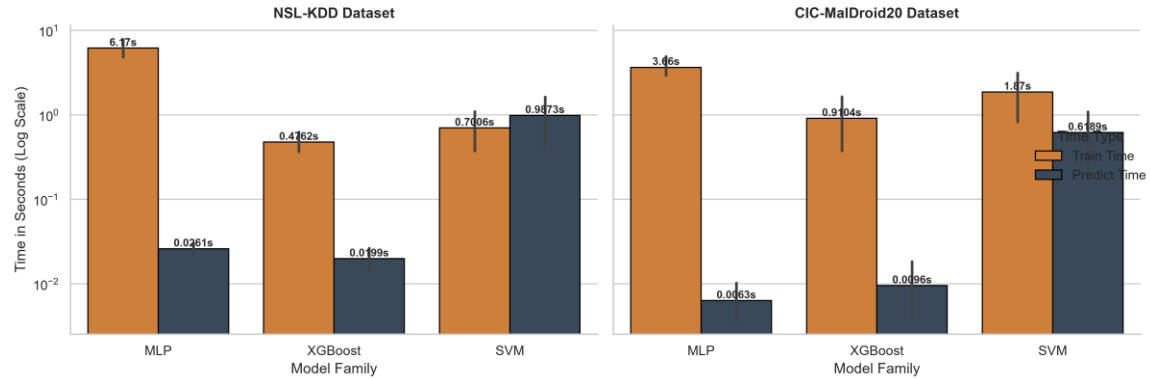
For CIC-MalDroid20, XGBoost is the absolute winner, achieving the highest Accuracy (95.04%) and Macro F1. For NSL-KDD, MLP (Optimized) narrowly took the lead in Accuracy (78.33%), but XGBoost remained highly competitive (77.86%). Overall, XGBoost is the most reliable, robust, and consistent model across both datasets.



Analysis:

By breaking down the GridSearchCV impact across all feature variants, a clear pattern emerges. MLP is highly dependent on tuning to achieve its best performance, showing massive gains (especially in CIC-MalDroid20). SVM shows moderate, consistent gains across variants. However, XGBoost shows almost zero improvement across all scenarios, and in some cases (negative values), tuning caused slight overfitting compared to its powerful default parameters.

Computational Cost: Train Time vs. Predict Time (Log Scale)

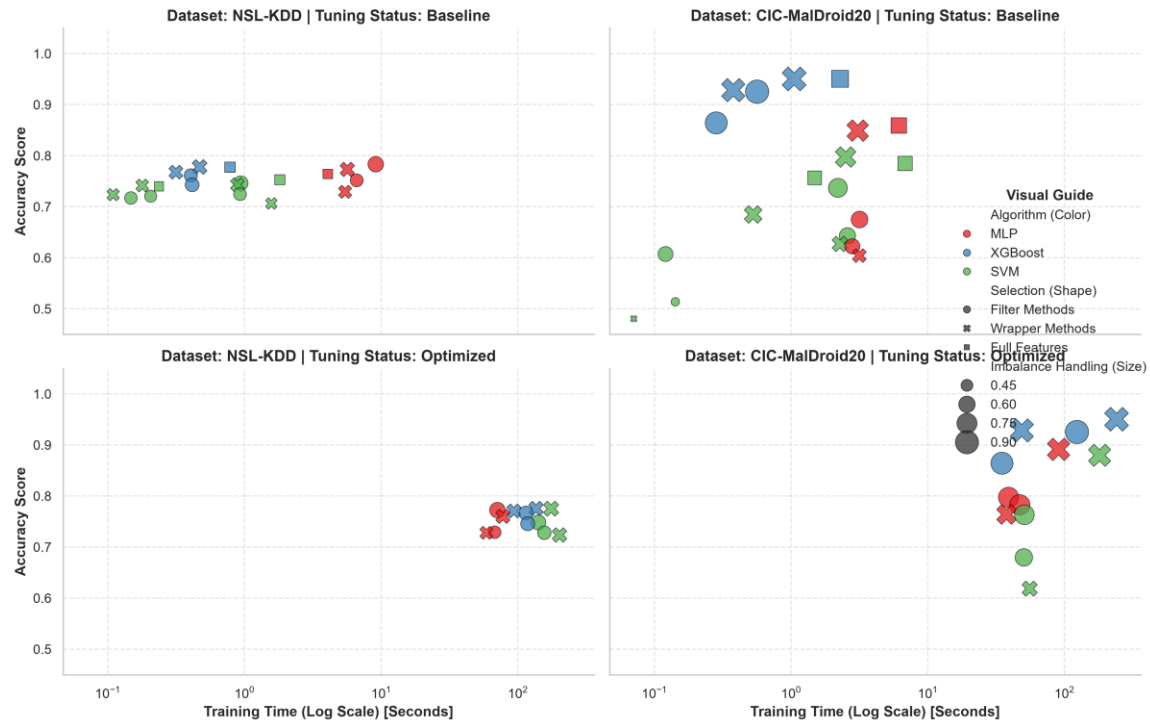


Analysis:

Analyzing the out-of-the-box (Baseline) models on a Logarithmic scale reveals a massive difference. MLP and SVM take significant Train Time, especially on the larger NSL-KDD dataset. XGBoost trains exceptionally fast. More importantly, for real-world Intrusion Detection Systems (IDS), 'Predict Time' is critical. XGBoost and MLP predict in mere milliseconds, making them highly suitable for real-time traffic analysis, whereas SVM is notably slower in prediction.

Bonus Analysis: The Ultimate Master Matrix (7-Dimensional View)

The Ultimate Master Matrix: Unveiling All 7 Dimensions Intuitively



Analysis:

This final 'Master Matrix' cleanly consolidates all 7 dimensions of the experiment:

1. Columns separate the Datasets (Left = NSL, Right = CIC).
2. Rows separate Baseline (Top) from Optimized/Tuned (Bottom) runs.
3. The X-Axis represents Computational Training Time on a log scale (further left is faster).
4. The Y-Axis represents the absolute Accuracy score (higher is better).
5. Color distinctly identifies the Model Family (e.g., Red = MLP, Blue = XGBoost).
6. Marker Shape represents the Feature Selection strategy (Circles = Full, Triangles = Wrapper, Squares = Filter).
7. Marker Size represents the Macro F1 score (Larger bubbles = Better handling of minority classes).

Key Visual Takeaways:

- CIC-MalDroid20 (Right column) shows dense, large bubbles clustering perfectly in the top-left quadrant (High Accuracy, High F1, Low Time), proving it is a high-quality dataset.
- NSL-KDD (Left column) models are scattered lower and feature much smaller bubbles, visually exposing their failure to handle the dataset's severe class imbalance despite having 'acceptable' accuracy.
- Comparing Top (Baseline) vs Bottom (Optimized) rows visually proves that tuning pushes points higher up (better accuracy) but drastically drags all points to the right (massive exponential increase in Train Time, especially for SVM and MLP).

